AI 101 @ Workplace

2026

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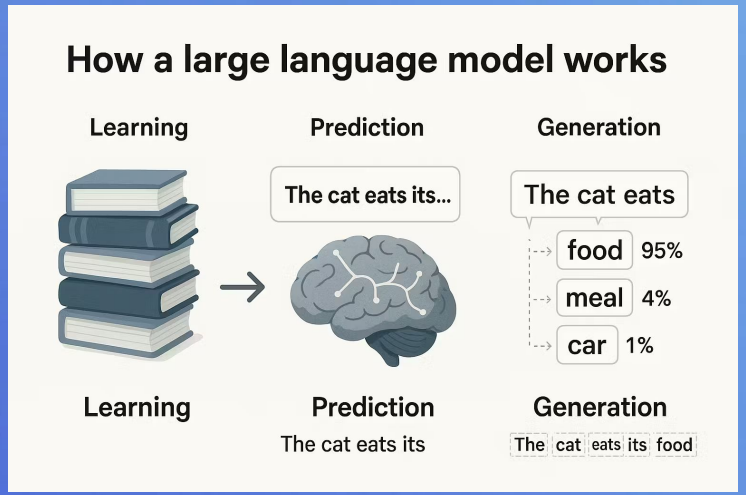
What happened?

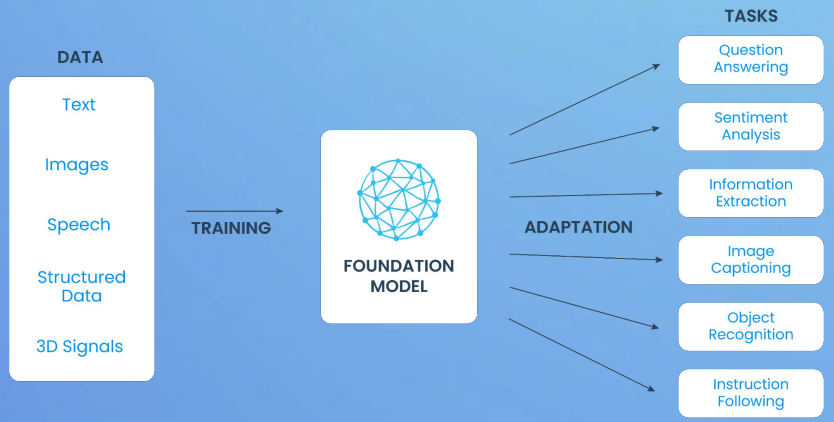
2017 - 2022: A new architecture ("Transformer") was developed which significantly accelerated the progress and development of AI (Artificial Intelligence).

November 2022: Release of ChatGPT by OpenAI, a public chatbot interface based on this technology.

2022-onwards: Increasing buzz, hype, and media attention surrounding LLM/AI tools and investments.

LLM: Large Language Model | **GPT**: Generative Pre-trained Transformer

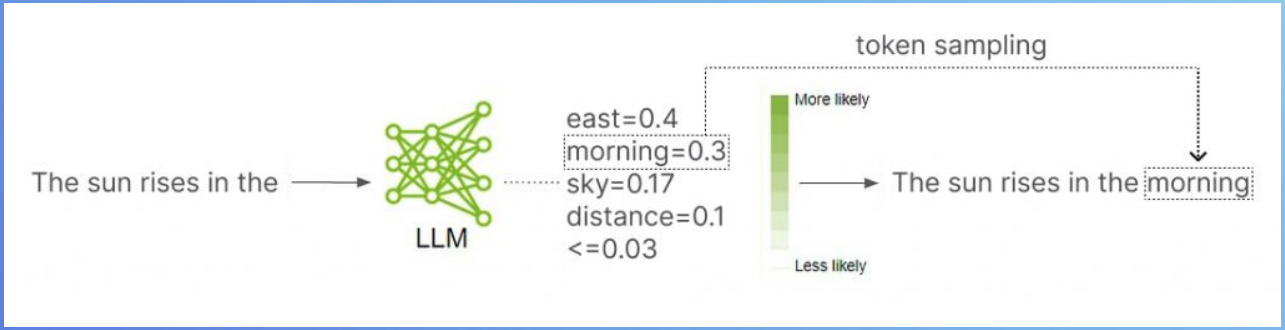




What happened?

Essentially, LLMs try to guess the most likely word in a sentence you provide. They “predict the next token”.

This is the “**G**” in Chat**G**PT. They can **generate** with significant accuracy!



This is possible because these models have been trained on an **extensive** and **exhaustive library of all human knowledge**, and have mapped out connections from it.

Why does this matter?

With these LLMs, you now can reasonably and reliably:

- Use them to discuss/explore how **you** can/should solve a problem (Guidance);
- Discuss/explore how **they** can/should solve the problem for you (Clarification/Planning);
- Have **them** solve the problem **directly** (Direct result);
- Have them **build the tools** you need to solve the problem **yourself** (Repeatable Solution).

Why does this matter?

This means that you can have **help** to do your **everyday work**!

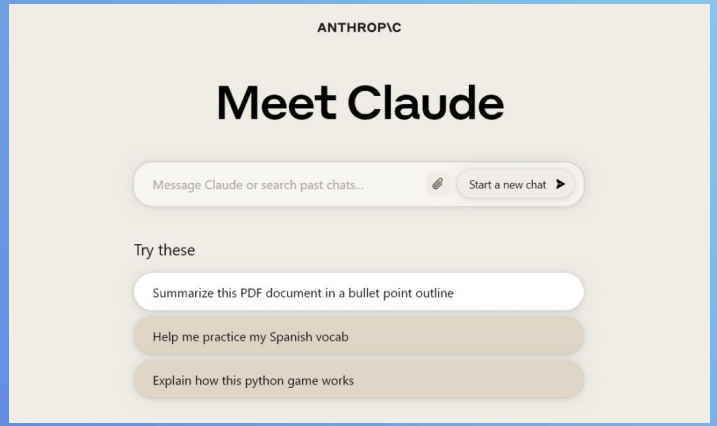
Anytime your work involves:

- Moving information/data (changing locations);
- Processing information/data (summarizing, elaborating, explaining);
- Translating information/data (changing languages, or converting from messy sources to organized structures;
- Finding information/data (on a document/file or the open web);
- Troubleshooting objects/machines/instruments/software.

You can use these tools to save time and effort!

How can we use it?

Several companies have released public models you can access via a chatbot interface.



Most of the companies in this field are either North American 🇺🇸 or Chinese 🇨🇳 .

How can we use it?

Some North American 🇺🇸 household names:

- ChatGPT by OpenAI (most recent model is GPT 5.3);
- Claude by Anthropic (most recent models are Opus/Sonnet/Haiku 4.6);
- Gemini by Google (most recent model is 3.1);
- Grok by X.AI (most recent model is 4.20);
- Llama by Meta (most recent model is Llama 4).

Some Chinese 🇨🇳 household names:

- Kimi by Moonshot AI (most recent model: Kimi K2.5);
- DeepSeek (most recent model: DeepSeek V3.2);
- GLM by Zhipu AI (most recent model: GLM-5);
- Qwen by Alibaba (most recent model: Qwen 3.5);
- MiniMax (most recent model: MiniMax M2.5).

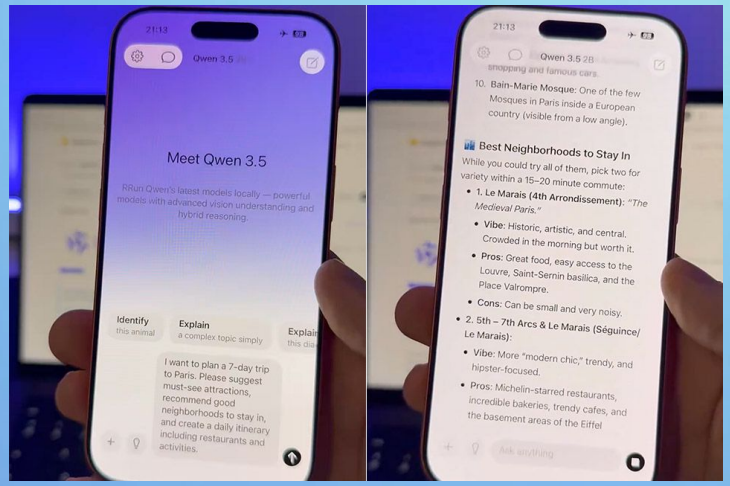
There is also one European 🇪🇺 household name: Le chat by Mistral AI (most recent model: Mistral Large 3).

How can we use it?

Most of these chatbots have free use with limitations, and then a subscription-based option for additional usage, more advanced models, and other quality of life features.

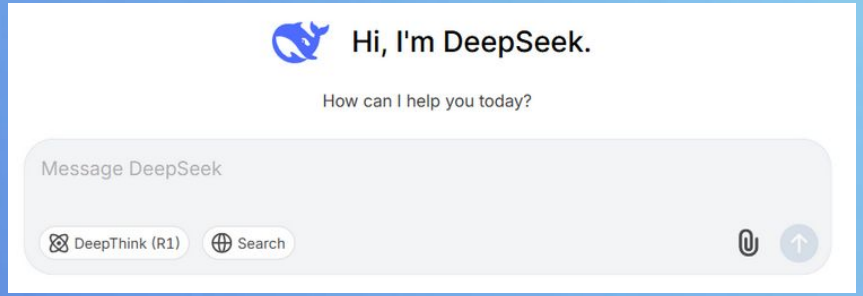
Some of these are also open-source and can be run locally, but as of today still require powerful expensive computers, effectively making them out of reach for the majority of the average consumers.

There are a few very small models that can be run in average consumer devices, and their capabilities improve steadily.



How <u>should</u> we use it?

For all these models, the main interface is via a chatbot where you type your request, query, or question. The practice of providing instructions is commonly known as "**prompting**".



LLMs work by “predicting the next token”. This means they are a **non-deterministic** engine (deterministic would mean the same input always results in the same output), but instead are a **probabilistic** engine (same input results in similar outputs but not necessarily equal), therefore there is a non-trivial amount of variation to be expected. As a result, this means they can also produce **mistakes**.

How <u>should</u> we use it?

Before you can take advantage of this technology and its potential at your workplace, you should:

1. Check which tools have been **vetted** for use by your **IT department**. There may be internal tools available for you;
2. Have **strong cognitive foundations** and **experience in your role and your field**, that allow you to determine what you need to do, how to do it, and when a task is finished or when it still needs refinement.

With a strong cognitive foundation, this allows you to begin using these AI tools safely and effectively. With guidance and practice directing AI with increasingly **clear specifications**, you can eventually graduate to handling autonomous AI systems with time, as your **judgement** develops.

How <u>should</u> we use it?

There are many examples and frameworks for good prompting, to help you achieve what you need. Here is one example , the **R.A.C.E. prompting framework**:

- **Role** example: "*You are a senior data analyst at our company. Explain the concept of data normalization to a new intern.*"
- **Action** example: "*Generate a list of 5 creative team-building activities for our remote team.*"
- **Context** example: "*We are a startup with a limited budget. Recommend free or low-cost tools for social media management.*"
- **Execute** example: "*Draft the email in a formal tone, and include a polite apology for the delay at the beginning.*"

Full Example Combining All 4 Categories: *"Act as a senior HR advisor at our tech company (**Role**). Draft a policy for remote work that encourages work-life balance (**Action**). Our team is global, so consider different time zones and cultural norms. The policy should be no longer than 2 pages and must include guidelines for synchronous and asynchronous communication (**Context**). Use a professional yet approachable tone, and include a section on mental health resources available to employees (**Execute**)."*

Where to begin?

Begin where the **friction** and **repetition** is at your work and your tasks. Don't try to automate your whole job at first! Start slow and safe, building a mental model of what works and what doesn't. **Protect the core and automate the edges**:

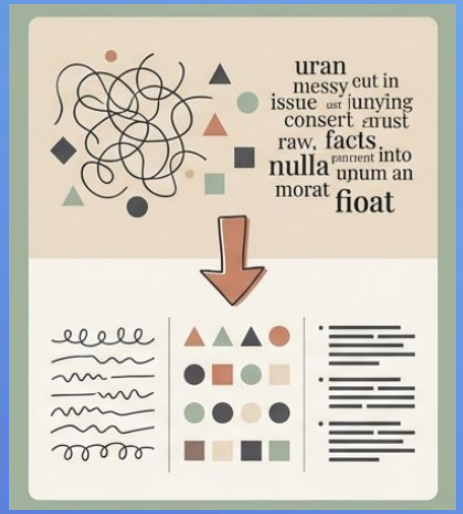
The **Core** (Human):

- Hard judgement, deep context, high-stakes decisions, relationship building, etc.

The **Edges** (AI):

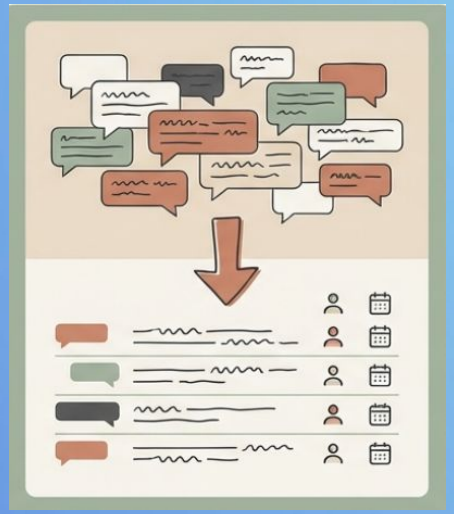
- Repetitive formatting, data extraction, initial synthesis, moving context from point A to point B, etc.

Where to begin?



Data preparation

Cleaning messy inputs, standardizing formats, and extracting raw facts from chaotic documents



Meeting synthesis

Consolidating sprawling discussion threads into clean summaries, identifying assigned owners, and tracking deadlines

Packaging & Formatting

Transforming rough analytical notes into structured deliverables

What to look out for?

AI models are confident, eager to please, and statistically prone to guessing. This can manifest as:

- **Hallucinations**: Models will invent plausible fictions to fill context gaps. Always verify outputs against ground truth.
- **False Confidence**: The AI does not know when it is wrong. It presents fabrications with the exact same tone as facts.
- **Privacy & Security Issues**: Never upload sensitive client data, PII (Personal Identifiable Information), or unreleased financials without explicit clearance.

What to look out for?

Since text/content is very easy to make now, there is a threat of “infinite **slop**”

Slop: Infinite, cheap, hollow content generated without human intervention. It may even be technically correct, but can be contextually useless.

The Trap: Reviewing slop does not scale. If you let AI generate unchecked volume, AI *will* generate unchecked volume, and you simply move your bottleneck from writing to editing.

The Fix: Use tested prompt frameworks, and force the AI to self-evaluate against strict constraints before you ever read the output.

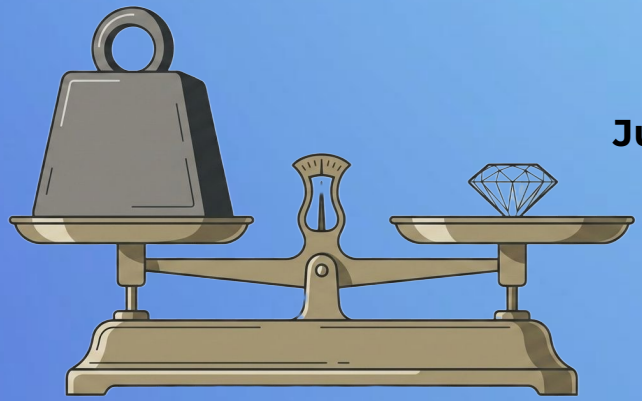
What to look out for?

Intelligence is now cheap, but **judgment** is rare.

AI can optimize perfectly toward a goal. Only humans can decide which goal is worth optimizing toward.

Good judgment is knowing what matters. It is distinguishing signal from noise, seeing second-order effects, and understanding when the rules apply and when they don't.

Computation



Judgment

What to look out for?

Your ultimate advantage is **taste**.

Taste is embodied, consequence-aware experience.

The AI has plenty of training data, but it has never lived through a failed product launch or a tense client meeting.

Taste is the **gut feeling** that an AI's output is technically perfect, but politically impossible or practically useless.

You are the editor/manager/orchestrator of a brilliant, alien intelligence. Your **strong cognitive foundations** and your **experience in your role and your field** have refined your taste and your instincts. Your taste is now the bridge between computation and reality.

What to look out for?

There is one last threat: over-reliance.

Being mindful of when these tools are **helping** vs when they are **replacing**.

Consider:

"*When is my **thinking** being **extended*** vs *when is my thinking being **atrophied**?*"

Each time you offload thinking without engaging with it first, you trade a small piece of future capability for present convenience.

What’s next?

Once you've had a few weeks/months of experience trying these tools, achieving goals, and experiencing mistakes, you are ready for **AI 201**